

E7 Solving trigonometric equations

Name: _____

Class: _____

Date: _____

Time: **225 minutes**

Marks: **186 marks**

Comments:

Q1.

- (a) Solve the equation $\cot x = 2$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(2)

- (b) Show that the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$ can be written as

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

(2)

- (c) Solve the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q2.

- (a) (i) Express $6 \sin \theta + 8 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give your value for α to the nearest 0.1° .

(2)

- (ii) Hence solve the equation $6 \sin 2x + 8 \cos 2x = 7$, giving all solutions to the nearest 0.1° in the interval $0^\circ < x < 360^\circ$.

(4)

- (b) (i) Prove the identity $\frac{\sin 2x}{1 - \cos 2x} = \frac{1}{\tan x}$.

(4)

- (ii) Hence solve the equation

$$\frac{\sin 2x}{1 - \cos 2x} = \tan x$$

giving all solutions in the interval $0^\circ < x < 360^\circ$.

(4)

(Total 14 marks)

Q3.

- (a) Solve the equation $\sin 2x = \sin 48^\circ$, giving the values of x in the interval $0^\circ \leq x < 360^\circ$.

(4)

- (b) Solve the equation $2 \sin \theta - 3 \cos \theta = 0$ in the interval $0^\circ \leq \theta < 360^\circ$, giving your answers to the nearest 0.1° .

(4)

(Total 8 marks)

Q4.

- (a) Solve the equation $\sec x = 3$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(3)

- (b) Show that the equation $\tan^2 x = 2 \sec x + 2$ can be written as $\sec^2 x - 2 \sec x - 3 = 0$.

(2)

- (c) Solve the equation $\tan^2 x = 2 \sec x + 2$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(4)

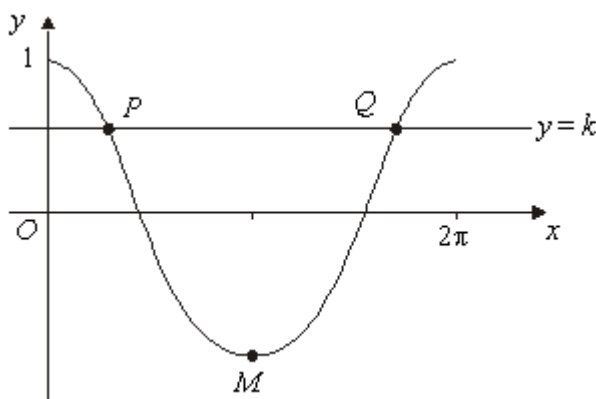
(Total 9 marks)

Q5.

- (a) Solve the equation $\cos x = 0.3$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures.

(3)

- (b) The diagram shows the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$, and the line $y = k$.



The line $y = k$ intersects the curve $y = \cos x$, $0 \leq x \leq 2\pi$, at the points P and Q .
The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of Q in terms of π and α .

(1)

- (c) Describe the geometrical transformation that maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$.

(2)

- (d) Solve the equation $\cos 2x = \cos \frac{4\pi}{5}$ in the interval $0 \leq x \leq 2\pi$, giving the values of x in terms of π .

(4)

(Total 12 marks)

Q6.

- (a) (i) Show that the equation

$$2 \cot^2 x + 5 \operatorname{cosec} x = 10$$

can be written in the form $2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$.

(2)

- (ii) Hence show that $\sin x = -\frac{1}{4}$ or $\sin x = \frac{2}{3}$.

(3)

- (b) Hence, or otherwise, solve the equation

$$2 \cot^2(\theta - 0.1) + 5 \operatorname{cosec}(\theta - 0.1) = 10$$

giving all values of θ in radians to two decimal places in the interval $-\pi < \theta < \pi$.

(3)

(Total 8 marks)

Q7.

- (a) Express $\cos 2x$ in terms of $\sin x$.

(1)

- (b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of x .

(2)

- (ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0^\circ < x < 360^\circ$.

(4)

- (c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

(2)

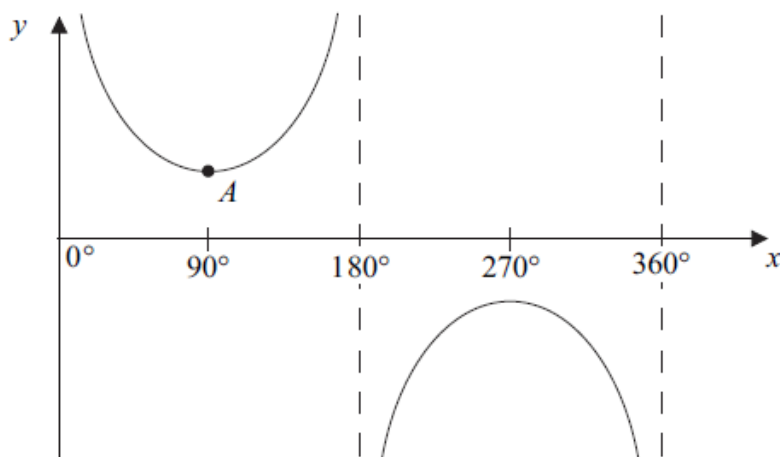
(Total 9 marks)

Q8.

- (a) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$ (3)
- (b) Write down the **two** solutions of the equation $\tan x = \tan 61^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$ (2)
- (c) (i) Given that $\sin \theta + \cos \theta = 0$, show that $\tan \theta = -1$. (1)
- (ii) Hence solve the equation $\sin (x - 20^\circ) + \cos (x - 20^\circ) = 0$ in the interval $0^\circ \leq x \leq 360^\circ$ (4)
- (d) Describe the single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan (x - 20^\circ)$. (2)
- (e) The curve $y = \tan x$ is stretched in the x -direction with scale factor $\frac{1}{4}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 13 marks)

Q9.

- (a) Solve the equation $\operatorname{cosec} x = 2$, giving all values of x in the interval $0^\circ < x < 360^\circ$. (2)
- (b) The diagram shows the graph of $y = \operatorname{cosec} x$ for $0^\circ < x < 360^\circ$.



Sketch the graph of $y = |\operatorname{cosec} x|$ for $0^\circ < x < 360^\circ$.

(2)
(Total 4 marks)

Q10.

- (a) Express $4 \cos x + 3 \sin x$ in the form $R \cos (x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 360^\circ$, giving your value for α to the nearest 0.1° . (3)
- (b) Hence solve the equation $4 \cos x + 3 \sin x = 2$ in the interval $0^\circ < x < 360^\circ$, giving all solutions to the nearest 0.1° . (4)
- (c) Write down the minimum value of $4 \cos x + 3 \sin x$ and find the value of x in the interval $0^\circ < x < 360^\circ$ at which this minimum value occurs. (3)
- (Total 10 marks)**

Q11.

- (a) Describe the geometrical transformation that maps the curve with equation $y = \sin x$ onto the curve with equation:
- (i) $y = 2 \sin x$; (2)
- (ii) $y = -\sin x$; (2)
- (iii) $y = \sin(x - 30^\circ)$. (2)
- (b) Solve the equation $\sin(\theta - 30^\circ) = 0.7$, giving your answers to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$. (3)
- (c) Prove that $(\cos x + \sin x)^2 + (\cos x - \sin x)^2 = 2$. (4)
- (Total 13 marks)**

Q12.

It is given that $2 \operatorname{cosec}^2 x = 5 - 5 \cot x$.

- (a) Show that the equation $2 \operatorname{cosec}^2 x = 5 - 5 \cot x$ can be written in the form

$$2 \cot^2 x + 5 \cot x - 3 = 0$$

(2)

- (b) Hence show that $\tan x = 2$ or $\tan x = -\frac{1}{3}$. (2)

- (c) Hence, or otherwise, solve the equation $2\operatorname{cosec}^2 x = 5 - 5 \cot x$, giving all values of x in radians to one decimal place in the interval $-\pi < x \leq \pi$.

(3)

(Total 7 marks)

Q13.

It is given that $3 \cos \theta - 2 \sin \theta = R \cos(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

- (a) Find the value of R .

(1)

- (b) Show that $\alpha \approx 33.7^\circ$.

(2)

- (c) Hence write down the maximum value of $3 \cos \theta - 2 \sin \theta$ and find a **positive** value of θ at which this maximum value occurs.

(3)

(Total 6 marks)

Q14.

- (a) Describe the single geometrical transformation by which the curve with equation $y = \tan \frac{1}{2}x$ can be obtained from the curve $y = \tan x$.

(2)

- (b) Solve the equation $\tan \frac{1}{2}x = 3$ in the interval $0 < x < 4\pi$, giving your answers in radians to three significant figures.

(4)

- (c) Solve the equation

$$\cos \theta (\sin \theta - 3 \cos \theta) = 0$$

in the interval $0 < \theta < 2\pi$, giving your answers in radians to three significant figures.

(5)

(Total 11 marks)

Q15.

- (a) Solve the equation $\sec x = 5$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(3)

- (b) Show that the equation $\tan^2 x = 3 \sec x + 9$ can be written as

$$\sec x^2 - 3 \sec x - 10 = 0$$

(2)

- (c) Solve the equation $\tan^2 x = 3 \sec x + 9$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 9 marks)

Q16.

- (a) (i) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$.

(1)

- (ii) Express $\cos 2x$ in terms of $\cos x$.

(1)

- (b) Show that

$$\sin 2x - \tan x = \tan x \cos 2x$$

for all values of x .

(3)

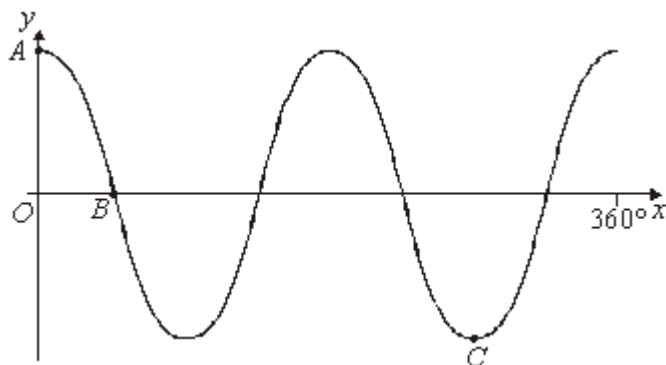
- (c) Solve the equation $\sin 2x - \tan x = 0$, giving all solutions in degrees in the interval $0^\circ < x < 360^\circ$.

(4)

(Total 9 marks)

Q17.

The diagram shows the graph of $y = \cos 2x$ for $0^\circ \leq x \leq 360^\circ$.



- (a) Write down the coordinates of the points A , B and C marked on the diagram.

(4)

- (b) Describe the single geometrical transformation by which the curve with equation $y = \cos 2x$ can be obtained from the curve with equation $y = \cos x$.

(2)

- (c) Solve the equation

$$\cos 2x = 0.37$$

giving all solutions to the nearest 0.1° in the interval $0^\circ \leq x \leq 360^\circ$.
 (No credit will be given for simply reading values from a graph.)

(5)

(Total 11 marks)

Q18.

- (a) (i) Show that the equation

$$4 \tan \theta \sin \theta = 15$$

can be written as

$$4 \sin^2 \theta = 15 \cos \theta$$

(1)

- (ii) Use an appropriate identity to show that the equation

$$4 \sin^2 \theta = 15 \cos \theta$$

can be written as

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

(2)

- (b) (i) Solve the equation $4c^2 + 15c - 4 = 0$.

(2)

- (ii) Hence explain why the only value of $\cos \theta$ which satisfies the equation

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

$$\text{is } \cos \theta = \frac{1}{4}.$$

(1)

- (iii) Hence solve the equation $4 \tan \theta \sin \theta = 15$ giving all solutions to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$.

(2)

- (c) **Write down** all the values of x in the interval $0^\circ \leq x \leq 90^\circ$ for which

$$4 \tan 4x \sin 4x = 15$$

giving your answers to the nearest degree.

(2)

(Total 10 marks)

Q19.

It is given that $\tan^2 x = \sec x + 11$.

- (a) Show that the equation $\tan^2 x = \sec x + 11$ can be written in the form

$$\sec^2 x - \sec x - 12 = 0$$

(2)

- (b) Hence show that $\cos x = \frac{1}{4}$ or $\cos x = -\frac{1}{3}$.

(3)

- (c) Hence, or otherwise, solve the equation $\tan^2 x = \sec x + 11$, giving all values of x to the nearest degree in the interval $0^\circ < x < 360^\circ$.

(3)

(Total 8 marks)

Q20.

- (a) Express $2 \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where R is a positive constant and α is an acute angle. Give your value of α to the nearest 0.1° .

(3)

- (b) Solve the equation $2 \sin x + \cos x = 1$ for $0^\circ \leq x < 360^\circ$.

(4)

(Total 7 marks)